Life Processes

The big idea

Living things keep themselves alive by carrying out **life processes**: nutrition, respiration, transportation and excretion. Even while you sleep, these processes keep repairing and maintaining your body.

Nutrition

- **Autotrophic**: organisms make their own food. Green plants use **photosynthesis**:
  $6CO_2 + 12H_2O \rightarrow$ (sunlight, chlorophyll) $\rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$
  Steps: chlorophyll absorbs light; light energy splits water into hydrogen and oxygen; CO_2 is reduced to carbohydrate. CO_2 enters through **stomata**, which open and close using **guard cells**.
- **Heterotrophic**: organisms depend on others for food. Amoeba engulfs food with **pseudopodia**; fungi break food down outside the body (saprophytic); some are parasites (Cuscuta, ticks).

HUMAN DIGESTION

Mouth: saliva has **salivary amylase**, which breaks starch into sugar.
Stomach: HCl (acidic medium, kills germs), **pepsin** (digests protein), mucus (protects the lining).
Small intestine: bile from the liver **emulsifies fats** and makes the medium alkaline; pancreatic juice has trypsin (proteins) and lipase (fats). **Villi** increase the surface area for absorption.
Large intestine: absorbs water; the anus removes waste.

Respiration

Glucose is broken down to release energy, stored as **ATP**.

- **Aerobic** (with oxygen, in mitochondria): glucose → CO_2 + water + lots of energy.
- **Anaerobic** in yeast: glucose → ethanol + CO_2 + energy (fermentation).
- **Lack of oxygen in muscles**: glucose → lactic acid + energy, which causes cramps.

The first step, glucose → pyruvate, happens in the **cytoplasm** in all cases.

Human respiratory path: nostrils → pharynx → larynx → trachea (rings of cartilage stop it collapsing) → bronchi → bronchioles → **alveoli**, where oxygen diffuses into blood. **Haemoglobin** in red blood cells carries oxygen.

Transportation

DOUBLE CIRCULATION

Blood passes through the heart **twice** in each cycle: right side → lungs → left side → body. The four chambers keep oxygenated and deoxygenated blood separate, giving efficient oxygen supply for warm-blooded animals.

- **Arteries** carry blood away from the heart (thick, elastic walls). **Veins** bring blood back (valves stop backflow). **Capillaries** are one cell thick for exchange.
- **Platelets** help blood clot. **Lymph** carries absorbed fat and drains extra fluid.
- In plants, **xylem** carries water and minerals up (pulled by **transpiration**); **phloem** carries food (sugar) both ways (**translocation**, uses ATP).

Excretion

The **kidneys** filter blood. Each kidney has about a million **nephrons**. In the nephron, the **glomerulus** (in Bowman's capsule) filters blood; useful substances

(glucose, amino acids, salts, water) are **reabsorbed** in the tubule; the rest becomes urine, collected in the bladder.

When kidneys fail, **dialysis** (artificial kidney) filters the blood. Plants excrete by removing oxygen and water through stomata, storing waste in old leaves and bark, and resins and gums.

Quick recap

- Nutrition: photosynthesis (plants), digestion with enzymes (humans).
- Respiration releases energy as ATP; aerobic or anaerobic.
- Heart: four chambers, double circulation; xylem and phloem in plants.
- Nephrons filter blood to form urine.

Practice questions

1. Bile juice helps in digestion by
   (a) digesting proteins (b) emulsifying fats (c) digesting starch (d) killing germs
2. The site of aerobic respiration in a cell is the
   (a) nucleus (b) cytoplasm (c) mitochondria (d) chloroplast
3. Which blood vessels have valves to prevent backflow?
   (a) Arteries (b) Veins (c) Capillaries (d) Alveoli
4. The functional unit of the kidney is the
   (a) neuron (b) alveolus (c) nephron (d) villus
5. Water moves up the xylem mainly because of
   (a) transpiration pull (b) gravity (c) phloem pressure (d) photosynthesis in roots
6. Lactic acid builds up in muscles during
   (a) aerobic respiration (b) anaerobic respiration due to lack of oxygen (c) photosynthesis (d) digestion

Answers: 1. b 2. c 3. b 4. c 5. a 6. b